t+3. Equation (4.11) also shows the origin of the name "indirect utility" because the indirect utility from the previous problem, at t+4, enters the direct utility from the current problem, at t+3.

Solving the problem two periods before the end gives us the optimal t+3 portfolio weight, x^*_{t+3}. We compute the maximum utility at t+3, which is the indirect utility at t+3:

$$V_{t+3} = E[U(1 + r^*_{p,t+4}(x^*_{t+3}))V_{t+4}]. \tag{4.12}$$

Panel C of Figure 4.2 shows the recursion applied once more to the t+2 problem having solved the t+3 and t+4 problems. Again, the t+2 optimization is a one-period problem. After solving the t+2 problem, we continue backward to t+1 and then finally to the beginning of the problem, time t. Dynamic programming turns the long-horizon problem into a series of one-period problems (following equation (4.11)). Dynamic programming is an extremely powerful technique, and Samuelson won the Nobel Prize in 1970 for introducing it into many areas of economics. Monetary policy (see chapter 9), capital investment by firms, taxation and fiscal policy, and option valuation are all examples of optimal control problems in economics that can be solved by dynamic programming. In continuous time, the value function is given by a solution to a partial differential equation called the Hamilton–Jacobi–Bellman equation. A more general form is called the Feynman–Kac theorem, widely used in thermodynamics. These are the same heavy-duty physics and mathematics concepts used in controlling airplanes and ballistic missiles. Portfolio choice is rocket science.

2.3. LONG-HORIZON INVESTING FALLACIES

The important lesson from the previous section on dynamic programming is not that you should hire a rocket scientist to do portfolio choice (although there are plenty of ex-rocket scientists working in this area), but that dynamic portfolio choice over long horizons is first and foremost about solving one-period portfolio choice problems. Viewing the dynamic programming solution of long-horizon portfolio choice this way demolishes two widely held misconceptions about long-horizon investing.

Buy and Hold Is Not Optimal

A long-horizon investor never buys and holds. The buy-and-hold problem is illustrated in Figure 4.3: the investor chooses portfolio weights at the beginning of the period and holds the assets without rebalancing over the entire long-horizon problem. The buy-and-hold problem treats the long-horizon problem as a single, static problem. Buy-and-hold problems are nested by the dynamic portfolios considered in the previous section; they are a special case where the investor's optimal choice is to do nothing. Buy and hold is dominated by the optimal dynamic